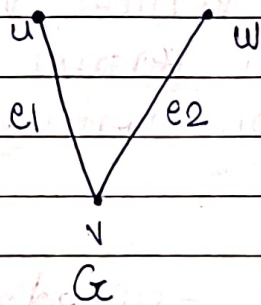


1.1 Graph

A Graph G is an ordered pair (V, E) where V is the non-empty set of vertices and E is the set of edges in which each element of E is assigned with an unique unordered pair of elements of V .

an element of set E is generally $e = (u, v)$ or $e = (v, u)$ where $u, v \in V$. here u & v are called end vertices of edge E

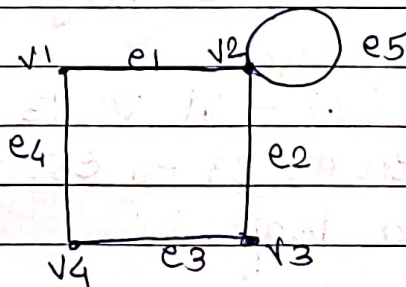
e.g.



Here

$$V = \{u, v, w\}$$

$$E = \{e1, e2\}$$



$$V = \{v1, v2, v3, v4\}$$

$$E = \{e1, e2, e3, e4, e5\}$$

*

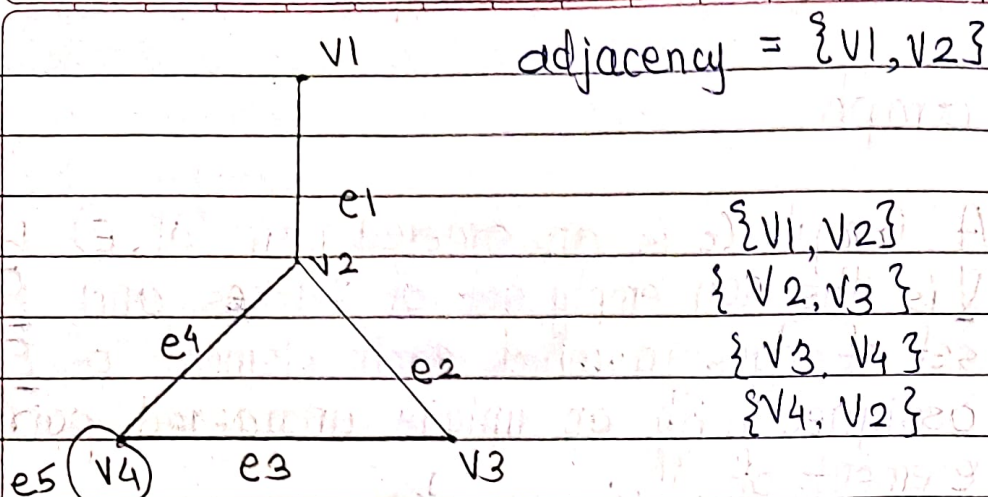
Adjacency.

Two vertices $(v1, v2)$ in a graph G are said to be adjacent to each other if and only if they are end vertices of the same edge (e)

*

Incidence.

if the vertex u is an end vertex of the edge e then the edge e is said to be incident on vertex u .



adjacent pair of vertices :-

edge e_1 is incident on vertices V_1 & V_2 ,
 edge e_2 is incident on vertices V_2 & V_3 ,
 edge e_3 is incident on vertices V_4 & V_3 ,
 edge e_4 is incident on vertices V_4 & V_2 .

* loop (self loop)

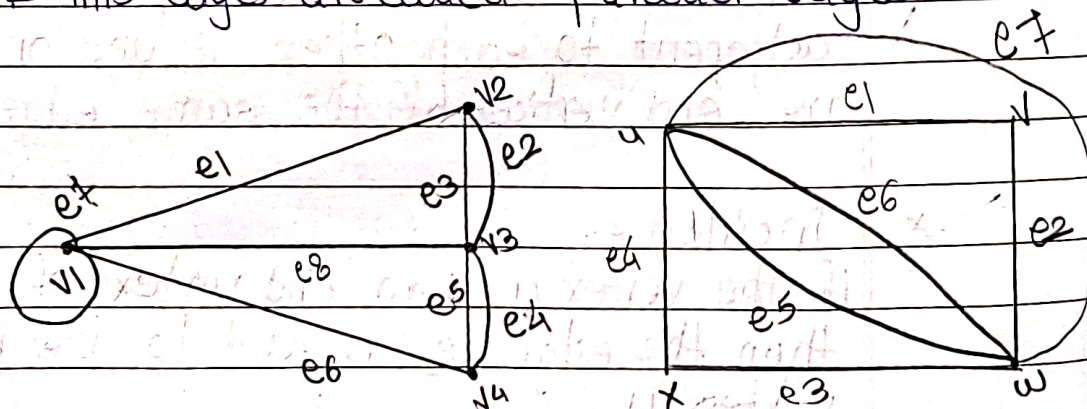
If both the end vertices of an edge are same then the edge is called loop.

here set of vertices $(V) = V_1, V_2, V_3, V_4$. and
 set of edges $E = e_1, e_2, e_3, e_4, e_5$.

hence edge e_5 is a loop.

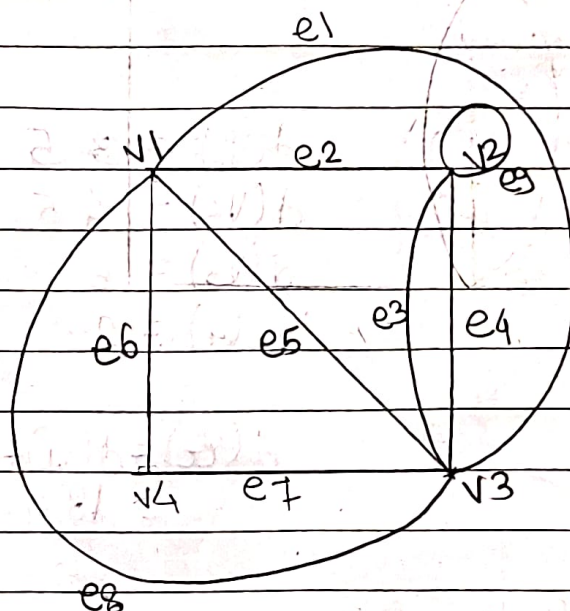
* Parallel Edges

if two or more edges have same terminal vertices then these edges are called parallel edges!



1. here e_2, e_3 are parallel edges also e_4 and e_5 are parallel edges.

2. here e_5, e_6 & e_7 are parallel edges

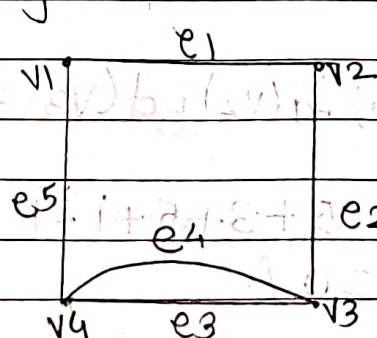


here, e_3, e_4 are parallel edges also e_1, e_8, e_5 are parallel edges. also e_9 is a loop.

* Degree of Vertex - 2m

The degree of vertex is defined as number of edges incident on it and it is denoted by $d(v)$.

For e.g.



$$d(v_1) = 2$$

$$d(v_2) = 2$$

$$d(v_3) = 3$$

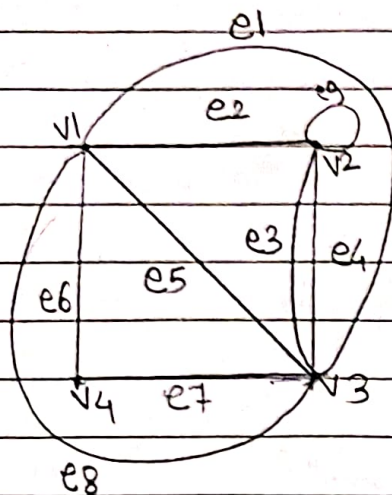
$$d(v_4) = 3$$

} vertex degree

$$d(G) = 10 \text{ } \} \text{ graph degree}$$

lx.

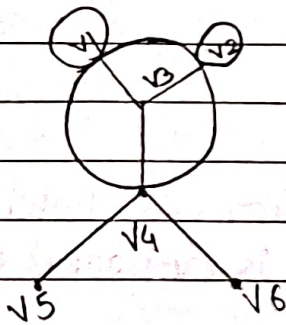
* Degree of the Graph $d(G)$:- - 4m
 Some sum of the degrees of all vertices in graph G is called degree of the graph (G) .
 it is denoted by $d(G)$ i.e. $\sum_{v_i \in G} d(v_i)$



$$\begin{aligned} d(v_1) &= 5 \\ d(v_2) &= 4 \\ d(v_3) &= 6 \\ d(v_4) &= 2 \end{aligned}$$

$$\begin{aligned} d(G) &= d(v_1) + d(v_2) + d(v_3) + d(v_4) \\ &= 18 \end{aligned}$$

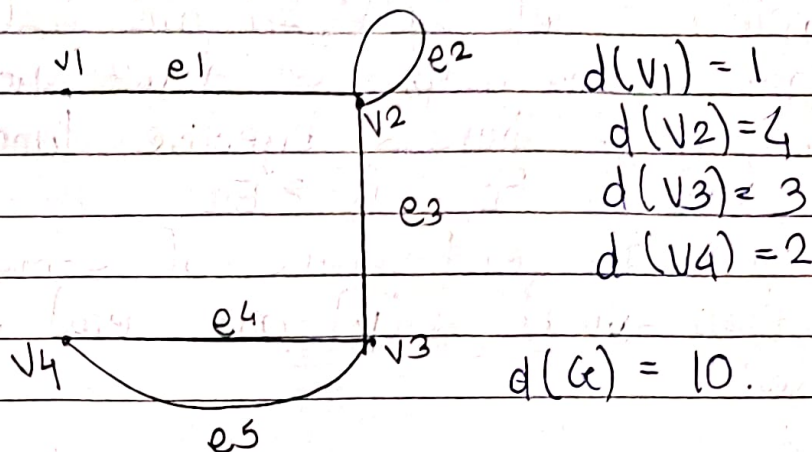
* 1. Find degree of the following graph



$$\begin{aligned} d(v_1) &= 5 \\ d(v_2) &= 5 \\ d(v_3) &= 3 \\ d(v_4) &= 5 \\ d(v_5) &= 1 \\ d(v_6) &= 1 \end{aligned}$$

$$\begin{aligned} d(G) &= d(v_1) + d(v_2) + d(v_3) + d(v_4) + d(v_5) + d(v_6) \\ &= 5 + 5 + 3 + 5 + 1 + 1 \\ d(G) &= 20 \end{aligned}$$

2.

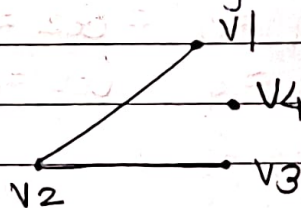


* Isolated Vertex.

A vertex with degree zero is called as isolated vertex

* Pendant Vertex.

A vertex with degree 1 is called as Pendant vertex.



here v_4 is isolated vertex

$$\therefore d(v_4) = 0$$

v_1 & v_3 are pendant vertex

$$\therefore d(v_1) \text{ \& } d(v_3) = 1$$



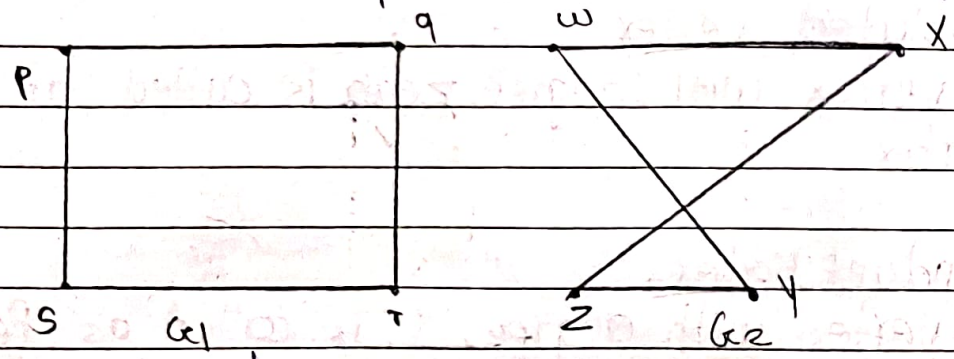
$\therefore v_4$ is isolated vertex

$\therefore v_1$ & v_3 are pendant.

imp

*** Isomorphism :-** let $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$ be two graphs. the graphs G_1 & G_2 are said to be isomorphic if there is bijective function $f_V : V_1 \rightarrow V_2$ & $f_E : E_1 \rightarrow E_2$ i.e if u, v are end vertices of some edge $e \in E_1$ then $f_V(u), f_V(v)$ are end vertices of $f_E(e)$

1. show that Graph G_1 & G_2 are isomorphic



→ No. of vertices in $G_1 = 4 = G_2 = 4$
No. of edges in $G_1 = 4$ $G_2 = 4$.

for Graph G_1

for Graph G_2

Vertex	degree	degree of adjacent Vertices	Vertex	degree	degree of adjacent vertices
p	2	2, 2	w	2	2, 2
q	2	2, 2	x	2	2, 2
r	2	2, 2	y	2	2, 2
s	2	2, 2	z	2	2, 2

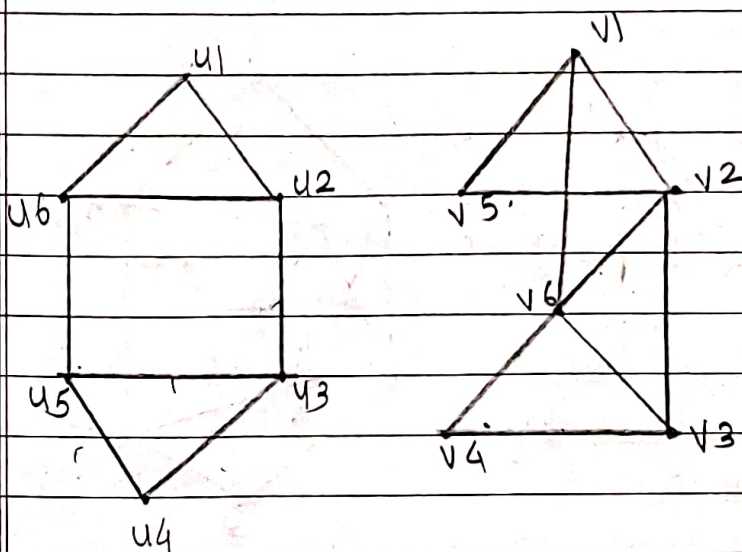
Solⁿ:- Here no. of vertices in $G_1 = 4$ &
no. of vertices in $G_2 = 4$
no. of edges in graph $G_1 = 4$ &
no. of edges in graph $G_2 = 4$.

isomorphic = $\cong$, $\neq$

Page No.			
Date			

$\therefore$ degree of adjacent vertices of graph G_1 =
 degree of adjacent vertices of G_2
 hence graph G_1 & G_2 are isomorphic.
 i.e. $G_1 \cong G_2$

2. determine if the following graphs are isomorphic.
 justify your answer.



$\rightarrow \therefore$ here no. of vertices in $G_1 = 6$
 no. of vertices in $G_2 = 6$
 no. of edges in graph $G_1 = 8$
 no. of edges in $G_2 = 10$.

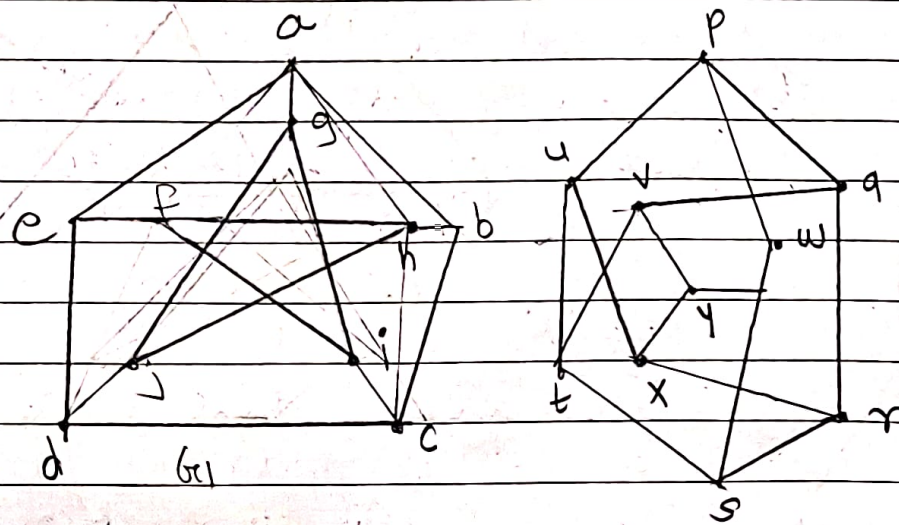
For Graph G_1

Vertex	degree	degree of adj. vertices	vertex	degree	D. of adj. v
u_1	2	3, 3	v_1	3	4, 2, 2
u_2	3	2, 3, 3	v_2	4	3, 3, 2, 4
u_3	3	3, 3, 2	v_3	3	4, 2, 4
u_4	2	3, 3	v_4	2	3, 4
u_5	3	2, 3, 3	v_5	2	3, 4
u_6	3	2, 3, 3	v_6	4	4, 3, 2, 4

$\therefore G_1 \neq G_2$.

Therefore degree of adj. vertices of graph G_1 is not equal to degree of adj. vertices of G_2 .
 $\therefore$ hence graph G_1 & G_2 are not isomorphic.
 $\therefore G_1 \not\cong G_2$

3. determine whether given graph G_1 & G_2 be isomorphic.



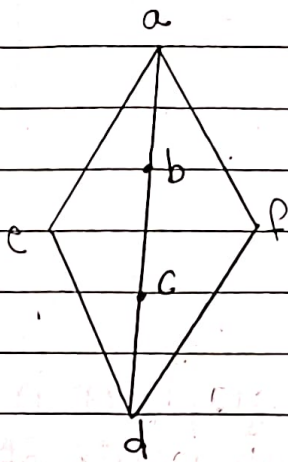
→ no. of vertices in $G_1 = 10$ G_2
 in $G_2 = 10$.
 no. of edges in G_1 & $G_2 = 15$.

for Graph G_1

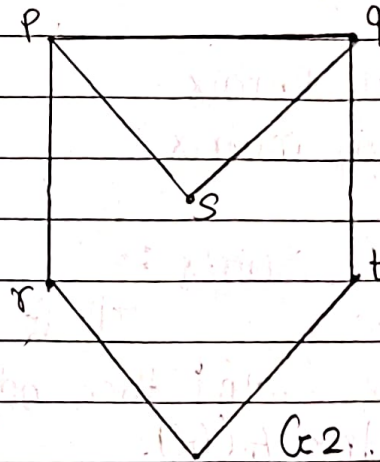
vertex degree d. of
 adj. vertex

a	3
b	3
c	3
d	3
e	3
f	3
g	3
h	
i	

4. determine whether following graph are isomorphic or not.



G₁



G₂

no. of vertices in $G_1 = 6$ in $G_2 = 6$

no. of edges in $G_1 = 7$, in $G_2 = 7$

Vertex	degree	d. of adj. vertex	Vertex	degree	d. of adj. vertex
a	3	2, 2, 2	p	3	3, 2, 2
b	2	3, 2	q	3	3, 2, 2
c	2	3, 2	r	2	3, 2
d	3	2, 2, 2	s	2	3, 3
e	2	3, 3	t	2	3, 2
f	2	3, 3	u	2	2, 2

from table 1 & 2 there is no one-one correspondence between graph G_1 & graph G_2 .

Therefore G_1 is not isomorphic to G_2 .

i.e. $G_1 \not\cong G_2$.

* Representation of Graph.

* Matrix Representation.

- i) Adjacency matrix
- ii) Incidency matrix.

i) Adjacency Matrix :-

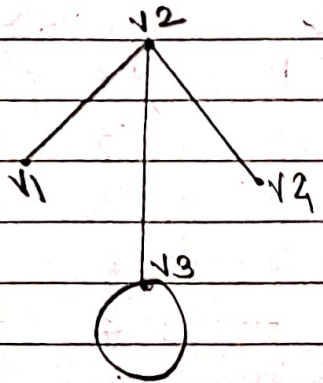
If G is a graph on n vertices set $\{v_1, v_2, \dots, v_n\}$ then adjacency matrix of G is the $n \times n$ matrix $A(G)$.

i.e. $A(G) = [a_{ij}]_{n \times n}$.

Where,

a_{ij} = no. of edges betⁿ v_i & v_j
 $= 2$ for self loop.

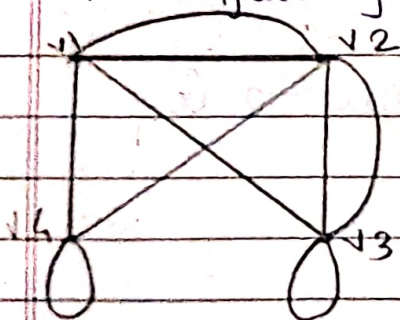
1. Consider the graph find adjacency matrix of graph



→

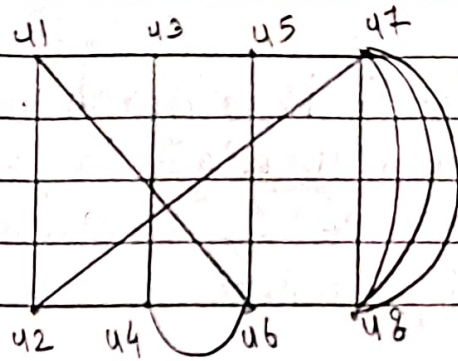
	v_1	v_2	v_3	v_4
v_1	0	1	0	0
v_2	1	0	1	1
v_3	0	1	2	0
v_4	0	1	0	0

2. Find adjacency matrix of following graph.



$A(G) =$

	v_1	v_2	v_3	v_4
v_1	0	2	1	1
v_2	2	0	2	1
v_3	1	2	2	0
v_4	1	1	0	2

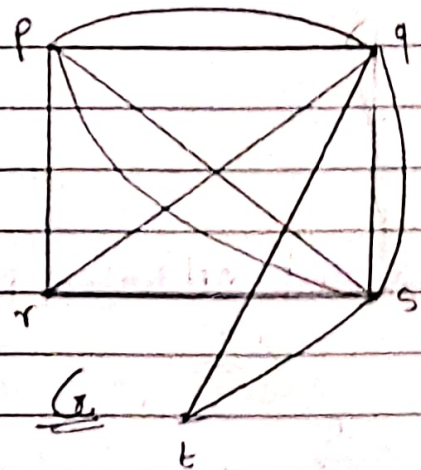


$A(G) =$

	u1	u2	u3	u4	u5	u6	u7	u8
u1	0	1	1	0	0	1	0	0
u2	1	0	0	1	0	0	0	0
u3	0	0	0	1	1	0	0	0
u4	0	1	1	0	0	2	0	0
u5	0	0	1	0	0	1	1	0
u6	1	0	0	2	1	0	0	1
u7	0	1	0	0	1	0	0	4
u8	0	0	0	0	0	1	4	0

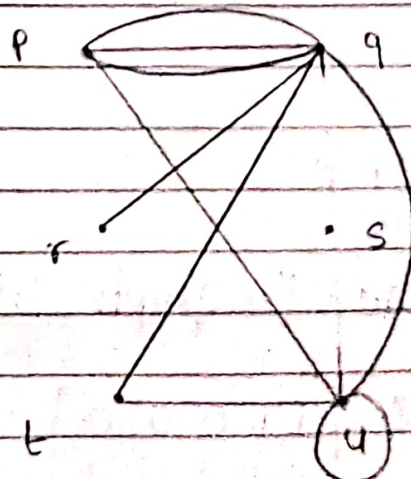
3. Draw a graph whose adjacency matrix is A where

$$A = \begin{bmatrix} 0 & 2 & 1 & 2 & 0 \\ 2 & 0 & 1 & 2 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$



4.

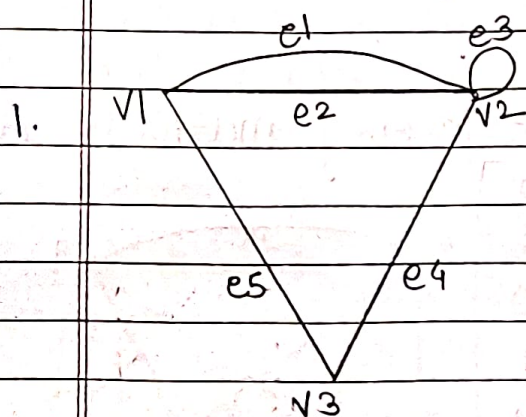
	p	q	r	s	t	u
p	0	3	0	0	0	1
q	3	0	1	0	1	1
r	0	1	0	0	0	0
s	0	0	0	0	0	0
t	0	1	0	0	0	1
u	1	1	0	0	0	2



* Incidence Matrix :-

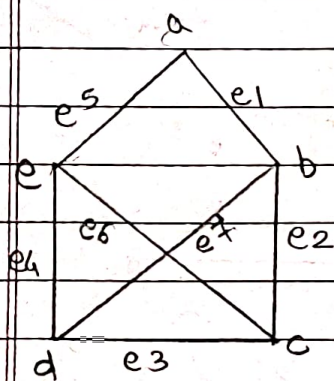
let G be a graph with n vertices $v_1, v_2, v_3 \dots v_n$ and m edges $e_1, e_2, e_3 \dots e_m$ then the incidence matrix of G is denoted by $I(G)$ as $I(G) = [a_{ij}]_{n \times m}$ where

$$a_{ij} = \begin{cases} 0 & \text{if } v_i \text{ is not incident with } e_j. \\ 1 & \text{if } v_i \text{ is incident with } e_j. \\ 2 & \text{if } e_j \text{ is a loop at } v_i. \end{cases}$$



$$I(G) = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

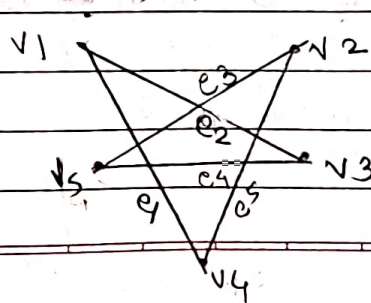
2. Find incidence matrices of the following graph



$$I(G) = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

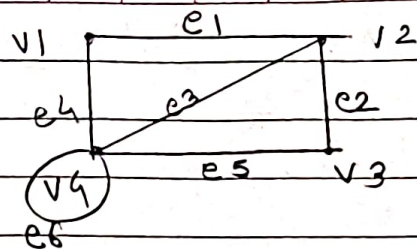
3. Draw the graph whose incidence matrix is given by

$$I = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$



4. $I(G) =$

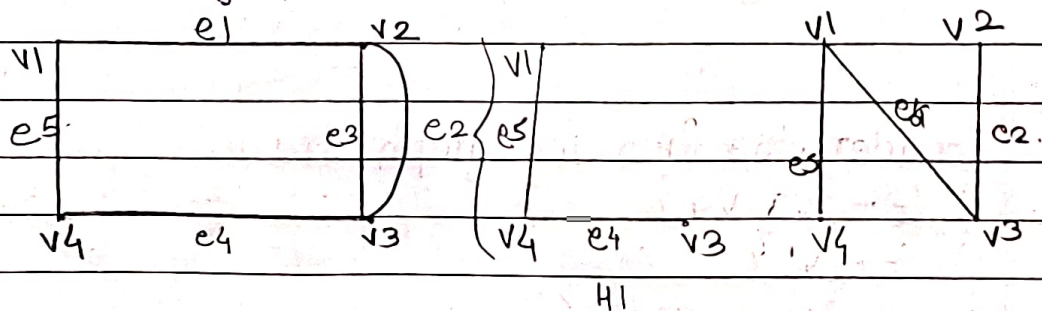
	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	0	0	1	0	0
v_2	1	1	1	0	0	0
v_3	0	1	0	0	1	0
v_4	0	0	1	1	1	2



* Subgraph.

A graph $H = (V(H), E(H))$ is called subgraph of $G = (V(G), E(G))$ if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.

1. Consider graph $G(V(G), E(G))$



$$V(H_1) \subseteq V(G) \quad V(H_2) = \{v_1, v_2, v_3, v_4\}$$

$$E(H_1) = \{e_4, e_5\} \quad E(H_2) = \{e_2, e_3, e_5\}$$

$$V(H_1) \subseteq V(G) \text{ \& } E(H_1) \subseteq E(G) \quad V(H_2) \subseteq V(G)$$

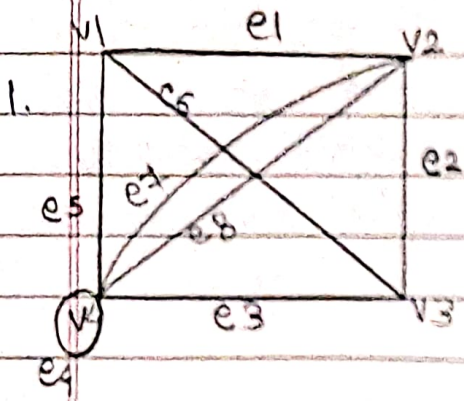
$$E(H_2) \not\subseteq E(G)$$

$\therefore H_2$ is not subgraph of G .

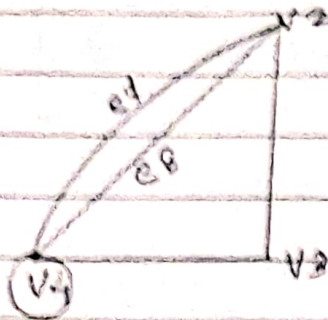
* Deletion of vertex.

(Vertex deleted subgraph).

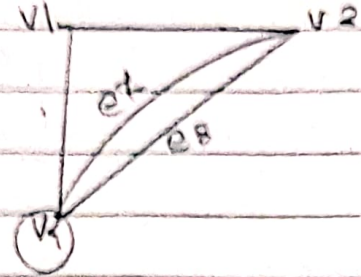
Let $G = (V(G), E(G))$ be a graph with at least two vertices & u be a vertex of G . The subgraph $G - u$ is obtained by deleting the vertex u from G together with all edges incident on u .



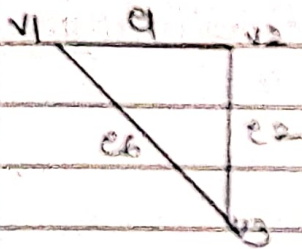
$G - v_1$



$G - v_3$



$G - v_4$



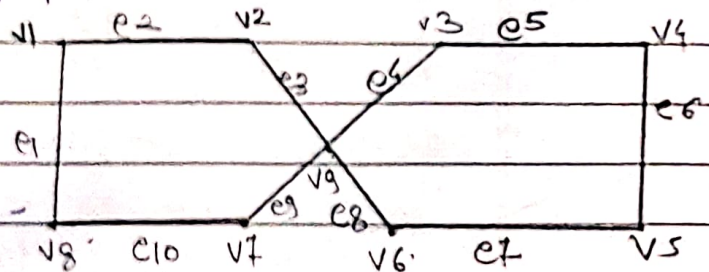
2. Consider the following graph draw

1) $G - \{v_3, v_4\}$

2) $G - \{v_1\}$

3) $G - \{v_1, v_2, v_7, v_8\}$

4) $G - \{v_1, v_2, v_9, v_3, v_4\}$



1) $G - \{v_3, v_4\}$

Walk :- let G be a graph, a finite alternating sequence $W: \{v_1, e_1, v_2, v_3, \dots, v_n\}$ of vertices and edges of G beginning and ending with vertex. Such that every edge in W is incident on vertex which precedes and succeeds it is called a Walk in G .

Remark :- 1. The walk is called $v_1 - v_n$

2. v_1 & v_n are called terminal vertices of the walk. and $v_2, v_3, \dots, v_{n-1}$ are called intermediate vertices.

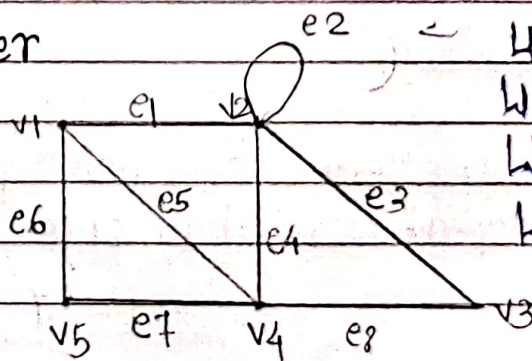
3. if terminal vertices are same in a walk then it is called as closed walk.

4. If terminal vertices are different then this walk are called as open walk. $\therefore$ open

5. In a walk and edge or vertex may be repeated more than once.

6. The number of edges in a walk (including repetition) is called length of the walk.

1. Consider



$W_1 = v_1$ to v_2 length is 2

$W_2 = v_2$ to v_3 length is 2

$W_3 = v_1$ to v_4 length is 2

$W_4 = v_5$ to e_3 length is 3

$W_5 = v_4$ to v_2 length is 3

$W_6 = v_3$ to v_5 length 2

→ $W_1 = v_1 e_1 v_2 e_2 v_2$

$W_6 = v_3 e_8 v_4 e_7 v_5$

$W_2 = v_2 e_2 v_2 e_3 v_3$

$W_7 = v_4 e_7 v_5 e_6 v_1$

$W_3 = v_1 e_6 v_5 e_7 v_4$

$W_8 = v_5 e_6 v_1 e_1 v_2 e_2 v_2$

$W_4 = v_5 e_7 v_4 e_8 v_3 e_3$

$W_9 = v_2 e_2 v_2 e_3 v_3 e_8$

$W_5 = v_4 e_8 v_3 e_3 v_2 e_2$

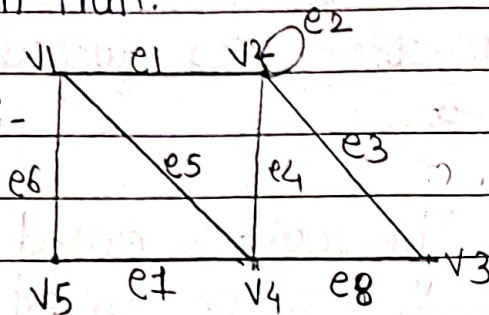
$W_{10} = v_1 e_1 v_2 e_2 v_2 e_3 v_3 e_8$

$v_4 e_5$

Trail :- A walk in which no edge is repeated

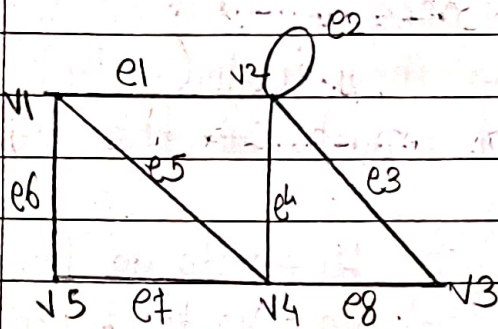
Note :- If ~~if~~ terminal vertices in a trail are same then it is called closed Trail. otherwise it is open trail.

example :-



1. $v_1 e_1 v_2 e_2 v_2 v_4$ it is trail.
2. $v_1 e_5 v_4 e_4 v_2 e_2 v_2 e_4 v_4$ it is not a trail
($\because e_4$ is twice)

Path :- an open walk in which no vertex is repeated more than once is called a Path.



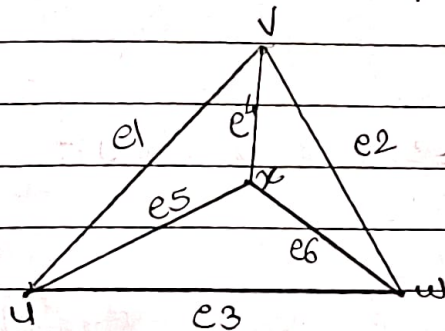
$e_4 v_4$ it is trail.

note :- Since every vertex does not repeat indirectly edge cannot repeat more than once. therefore every path is trail.

2. loop cannot be included in a path
3. the number of edges in a path is called length of the path.

4. any two path with same number of vertices are isomorphic

1. Find all u to v path in the following graph G . -4m



$$P_1 = u e_1 v$$

$$P_2 = u e_5 x e_4 v$$

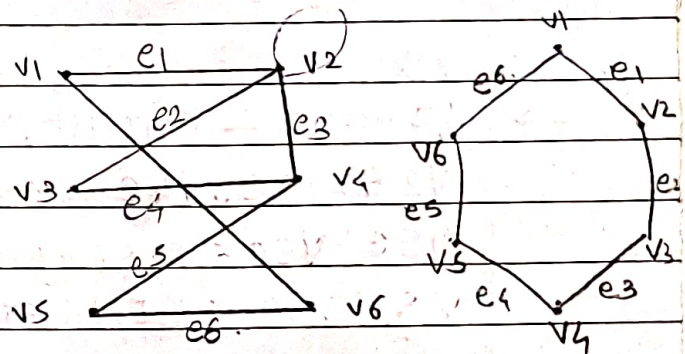
$$P_3 = u e_3 w e_6 x e_4 v$$

$$P_4 = u e_3 w e_2 v$$

$$P_5 = u e_5 x e_6 w e_2 v$$

2. Write all v_1-v_4 paths in a graph G having adjacency matrix.

	v_1	v_2	v_3	v_4	v_5	v_6
v_1	0	1	0	0	0	1
v_2	1	0	1	0	0	0
v_3	0	1	0	1	0	0
v_4	0	0	1	0	1	0
v_5	0	0	0	1	0	1
v_6	1	0	0	0	1	0



$$P_1 = v_1 e_1 v_2 e_3 v_4$$

$$P_2 = v_1 e_6 v_6 e_5 v_5 e_4 v_4$$

$$P_3 =$$

* Cycle (Circuit)

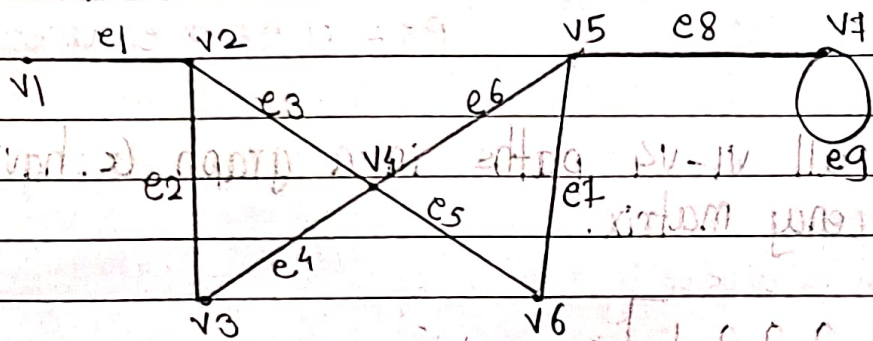
A closed walk in which no vertex (except initial and final) appears more than once is called a

cycle or circuit.

Note:- Self loop is a circuit

regular graph of degree 2 is a circuit.

1. Consider



→ $v_7 e_8 v_7$ - circuit, length = 1

$v_2 e_2 v_3 e_4 v_4 e_3 v_2$ - circuit, length = 3

$v_5 e_7 v_6 e_5 v_4 e_6 v_5$ - circuit, length = 3

$v_1 e_1 v_2 e_3 v_4 e_5 v_6 e_7 v_5 e_8 v_7$ - It is path, length = 8.

* Connected Graph.

A graph G is said to be connected if there is at least one path between every pair of vertices in G . Otherwise G is said to be disconnected graph.

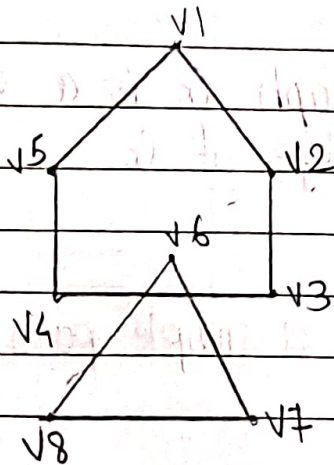
Therefore a graph G is said to be disconnected if there is a pair of vertices in G such that there is no path between them.

unit - II

Page No.

Date

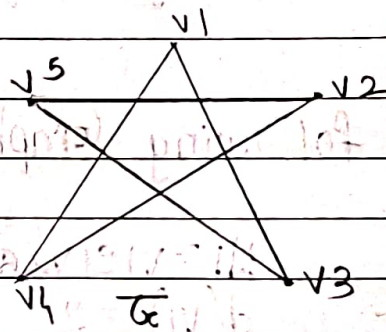
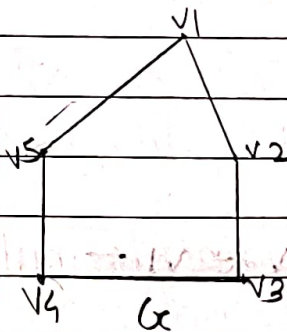
1.



$\Rightarrow$ It is disconnected graph
Since there is no path between
vertices betⁿ v_6 & v_3

2.

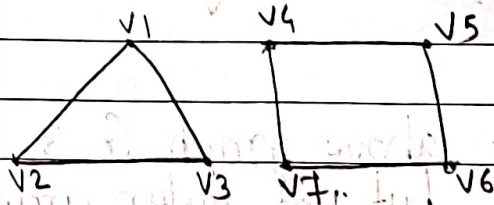
determine true or false. "If G is connected then
 G path $\bar{G}$ (compliment of G) then it's
if G is disconnected.



$\therefore$ Given statement
is false.

3. determine statment is True or false "every disconnected
graph has an isolated vertex.

$\Rightarrow$



* Euler's Graph

An Eulerian circuit in a graph G is a simple circuit containing every edge of G .

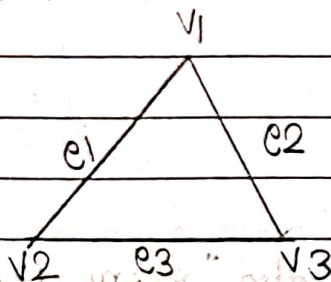
* Eulerian Path

An Eulerian path in a G is a simple path containing every edge of G .

* Euler Graph

If some close walk (W) in a graph G contains all the edges of the graph G the W is called an Euler Path & the given graph G is called Euler Graph.

1. Consider the following Graph.



$W_1 = v_1 e_1 v_2 e_3 v_3 e_2 v_1 \leftarrow$ Euler circuit.

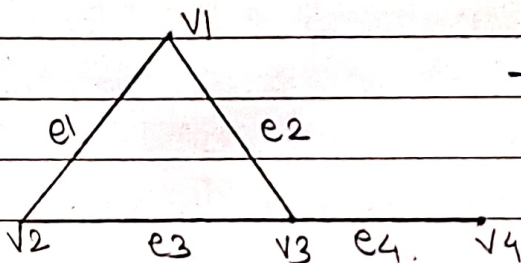
$$d(v_1) = 2$$

$$d(v_2) = 2$$

$$d(v_3) = 2$$

Note :- every Euler graph is always connected, but converse is not true.

2.

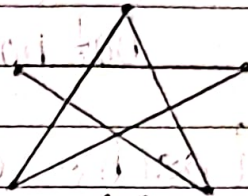


- above graph G is connected but not Euler graph

3. The following graph are euler graph.

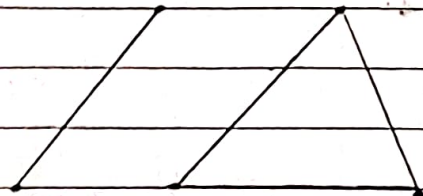


G_1



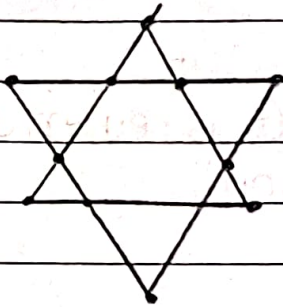
G_2

→ eulerian path



G_3

G_4



Imp

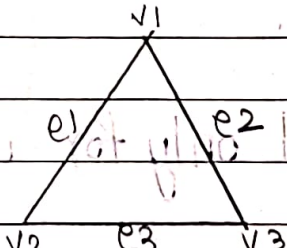
Thm:- The Connected graph G is an eulerian graph ~~if~~ if and only iff all vertices of G are of even degree.

* Hamiltonian Graph

* Hamiltonian Circuit - is a circuit which passes through each vertex exactly once. except initial vertex.

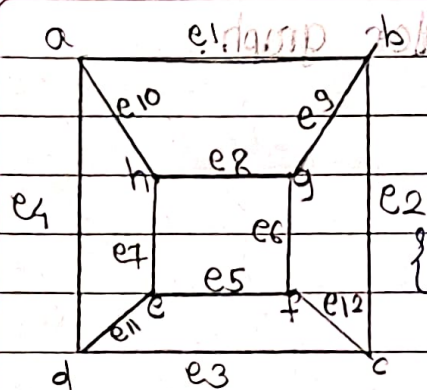
* Hamiltonian Graph :- a graph G is said to be hamiltonian graph if G has hamiltonian circuit.

1.



G_1

2.



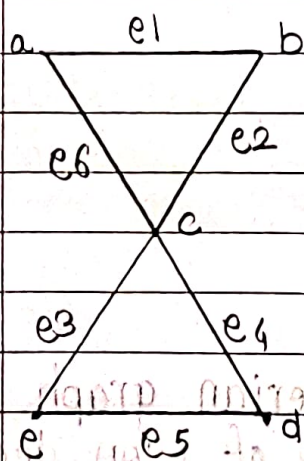
= above graph G is hamiltonian but not eulerian.

{ a e1 b e2 c e3 d e4 e5 e6 f e7 g e8 h e9 i e10 j }

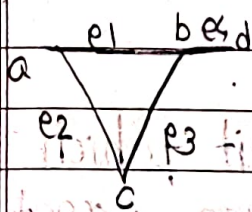
Hamiltonian Circuit.

g.

3.



- above graph G is eulerian but not hamiltonian.



Given graph G is not eulerian & not hamiltonian.

Remark:- 1. There are no relation between hamiltonian graph and eulerian graph.

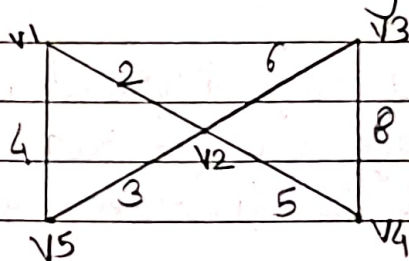
2. hamiltonian circuit in a graph of n vertices consists of exactly n edges.

3. hamiltonian circuit is defined only for connected simple graph.

* Fleury's algorithm :- using this algorithm we can find eulerian circuit. as follows.

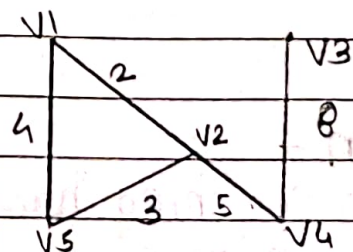
1. choose a vertex in graph G .
2. start ~~not~~ walking along the edges.
3. erase the edge traversed.
4. erase the isolated vertex.
5. keep walking until the initial vertex is reached by traversing all edges and crossing all vertices.
6. The value of circuit or walk is the sum of all edges during the walk.

1. using fleury's algorithm find an eulerian circuit in the following graph —4/6

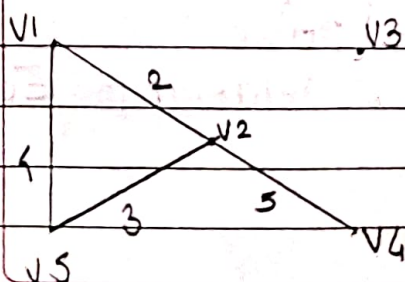


let us start with v_2 .
so we use

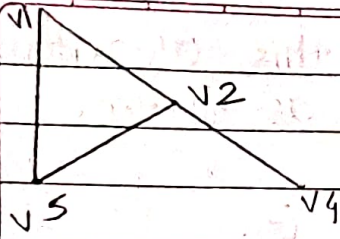
→ by fleury's algorithm let us start with vertex v_2 & with travel towards v_3 . Therefore the edge traversed is 6. Therefore erase this edge.



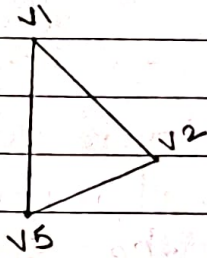
now we are at v_3 . we travel towards v_4 . Therefore edge traversed is 8. Therefore erase this edge.



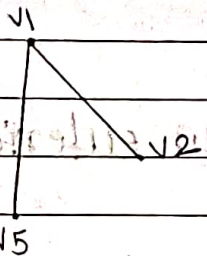
vertex v_3 is isolated vertex. therefore erase vertex v_3



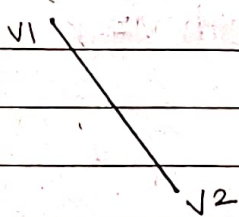
we are at V_4 and we travel towards V_2 and the edge traversed is 5 therefore erase this edge and also V_4 is isolated vertex erase V_4 .



we are at V_2 & edge traversed is 3. erase this edge



V_5 is isolated vertex we travel towards V_1 so edge traversed is 4 and V_5 is isolated vertex we delete the vertex V_5



from V_1 & V_2 edge traversed is 2. so erase this edge we get two isolated vertex V_1 & V_2 . therefore erase this vertex.

$V_2 \xrightarrow{6} V_3 \xrightarrow{8} V_4 \xrightarrow{5} V_2 \xrightarrow{3} V_5 \xrightarrow{4} V_1 \xrightarrow{2} V_2$

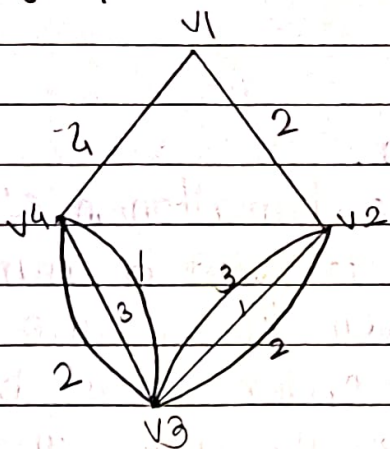
Therefore eulerian circuit is,

* Chinese postman problem -

In ancient time postman in china had to ride on a bicycle to collect the letters from postboxes and had to return to head office

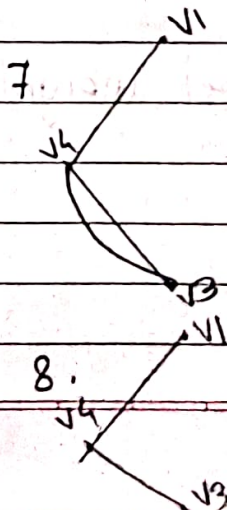
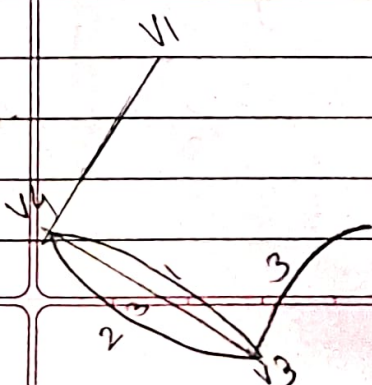
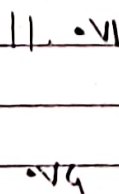
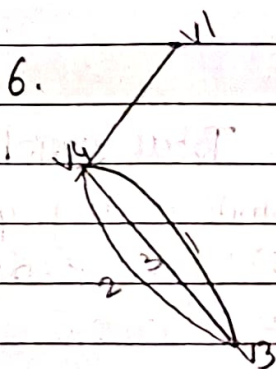
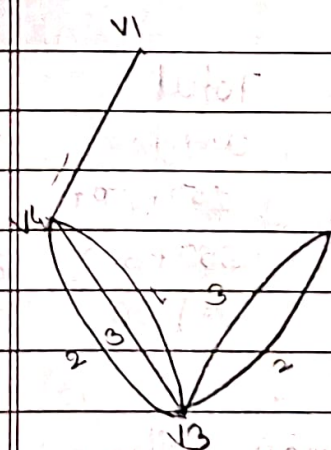
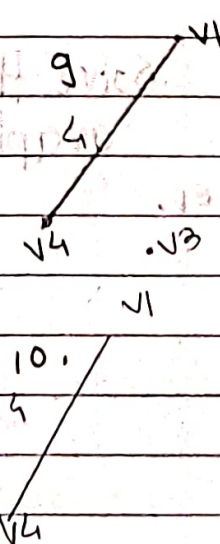
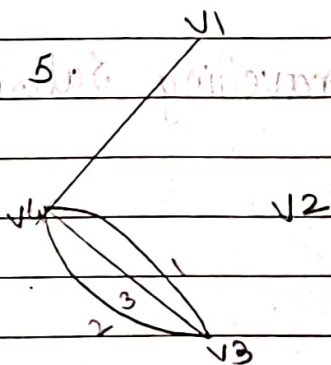
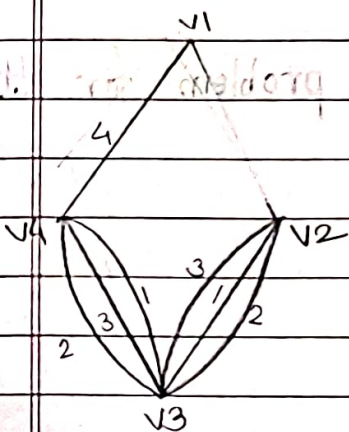
Chinese postman problem is related to Euler's graph.

1. Solve the chinese postman problem for the given graph below.



using chinese postman problem we find ^{shortest} ~~vertex~~ eulerian circuit containing all the edges of given connected graph as follows:

start from v_1 traversed edge to & go to v_2



12. v_1

eulerian circuit is
 $v_1 \xrightarrow{2} v_2 \xrightarrow{2} v_3 \xrightarrow{2} v_4 \xrightarrow{2} v_1$
 $\xrightarrow{2} v_1 \xrightarrow{2} v_2 \xrightarrow{2} v_3 \xrightarrow{2} v_4 \xrightarrow{2} v_1$

David $N_1 \xrightarrow{2} V_2 \xrightarrow{1} V_3 \xrightarrow{2} V_2 \xrightarrow{3} V_3 \xrightarrow{1} V_4 \xrightarrow{2} V_3 \xrightarrow{3} V_4 \xrightarrow{1} V_1$

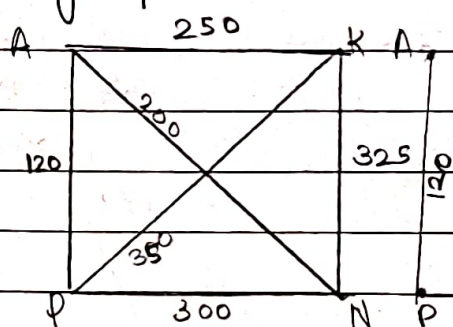
shortest eulerian circuit is $= 18$.

2+

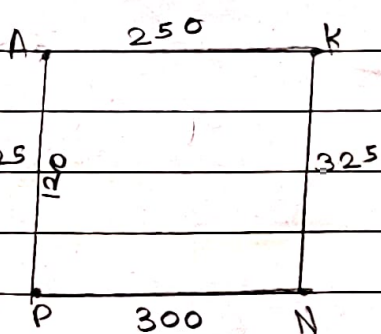
* Travelling Salesman Problem.

This problem is related to hamiltonian circuit. a salesman is required to visit a number of cities during a trip. given the distance between cities in what order should be travelled so as visit every city exactly once and return home with minimum distance travelled.

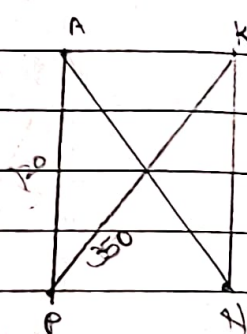
Solve the travelling Salesman problem for the foll. graph.



Total weight of
hamiltonian graph =
 $250 + 120 + 325 + 300$
 $= 995$

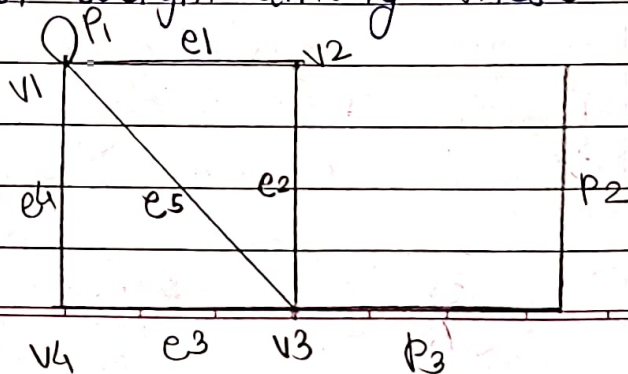


Total
weight =
 $250 + 200 +$
 $350 + 300$
 $= 1100$



Total
weight =
 $120 + 350 +$
 $325 + 200$
 $= 995$

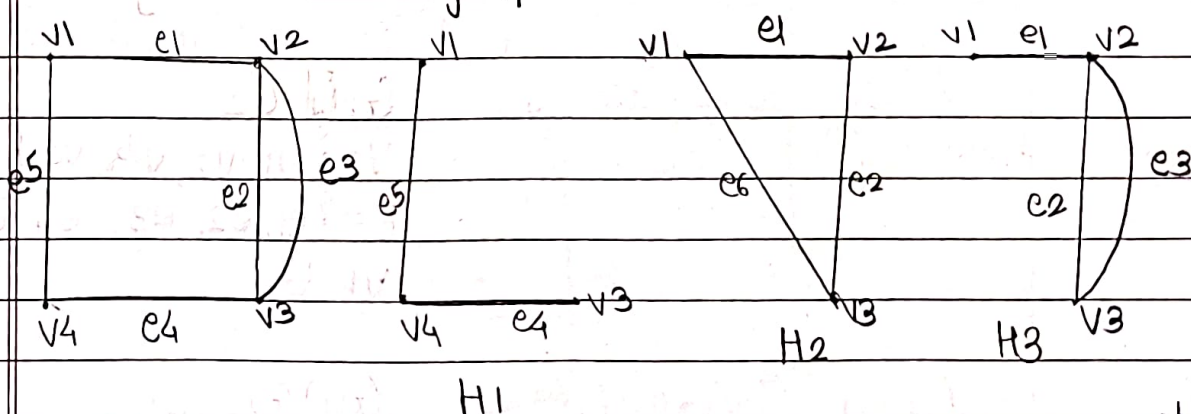
$\therefore$ hence the smallest weight among these hamiltonian circuit is 995.



* Operation on Graph.

* Subgraphs — a graph $(H) = (V(H), E(H))$ is called subgraph of $G = (V(G), E(G))$ if $V(H) \subseteq V(G)$ & $E(H) \subseteq E(G)$

For ex: Consider a graph.



$\therefore$ Here H_1 is subgraph of G
 $\therefore V(H_1) \subseteq V(G)$ & $E(H_1) \subseteq E(G)$.

$\therefore$ Here H_2 is not subgraph of G .

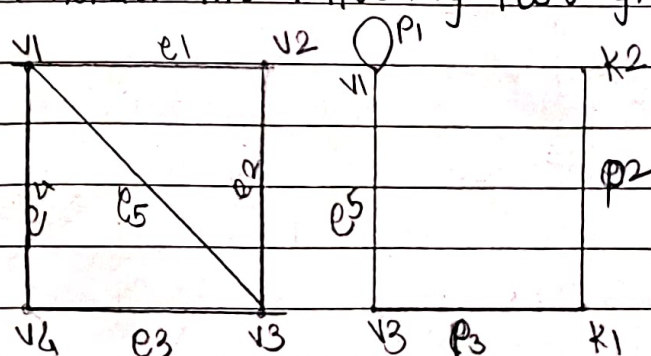
$\therefore V(H_2) \subseteq V(G)$ but $E(H_2) \not\subseteq E(G)$.

$\therefore H_3$ is subgraph of G .

* Deletion of vertex (Vertex deleted subgraph)

* Union — let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs union of G_1 & G_2 is a graph whose vertex set is $V = V_1 \cup V_2$ & edge set is $E = E_1 \cup E_2$

1. Consider the following two graph



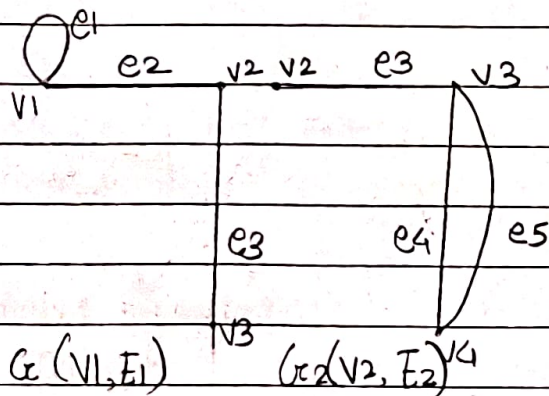
$G_1(V_1, E_1)$

$G_2(V_2, E_2)$

$G_1 \cup G_2$
 $V = V_1 \cup V_2$
 $= \{v_1, v_2, v_3, v_4, k_1, k_2\}$
 $E = E_1 \cup E_2$
 $\{e_1, e_2, e_3, e_4, e_5, p_1, p_2, p_3\}$

* Intersection :- let $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$ be two graphs. Intersection of G_1 & G_2 is a graph whose vertex set is $V = V_1 \cap V_2$ & the edge set is $E = E_1 \cap E_2$

1. find $G_1 \cup G_2$ and $G_1 \cap G_2$ of following graph.



$G_1 \cap G_2$

$$V = \{v_1, v_2, v_3, v_4\} = V_1 \cup V_2$$

$$E = \{e_1, e_2, e_3, e_4, e_5\} = E_1 \cup E_2$$

$$V_1 = V_2$$

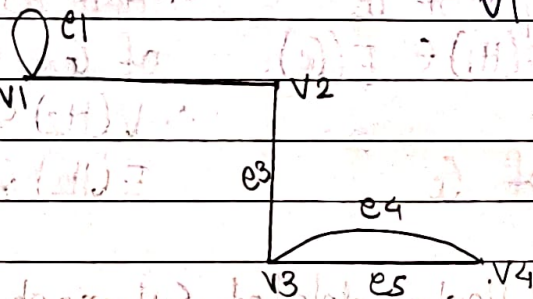
$G_1(V_1, E_1)$

$G_2(V_2, E_2)$

$G_1 \cap G_2$

$$V_1 \cap V_2 = V = \{v_2, v_3\}$$

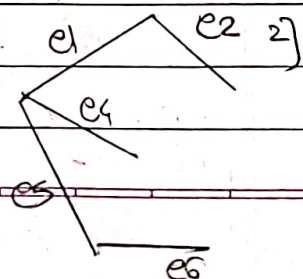
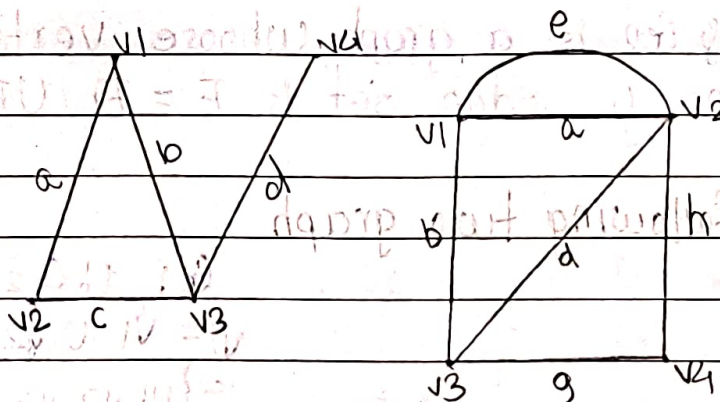
$$E = \{e_3\}$$

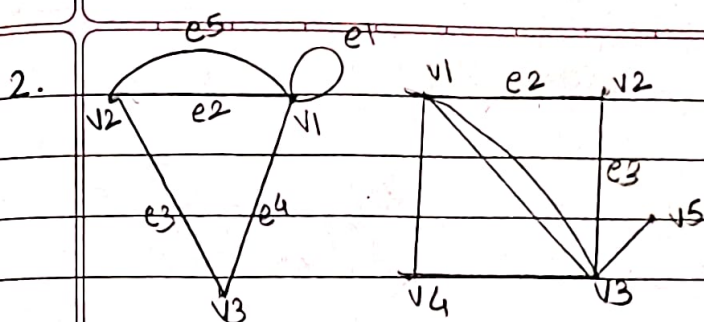


v_2 e_3 v_3

2. find union and intersection of the following graph.

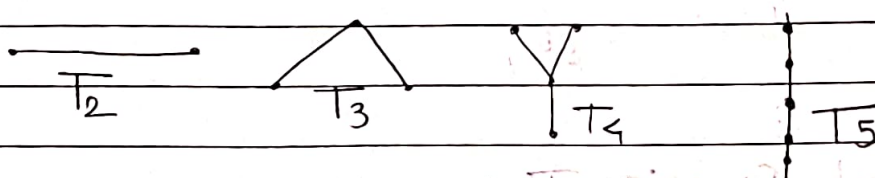
1.





* Tree - A connected graph without any cycle (circuit) is called tree.

notation - tree graph on n vertices denoted as T_n



Properties of Tree graph

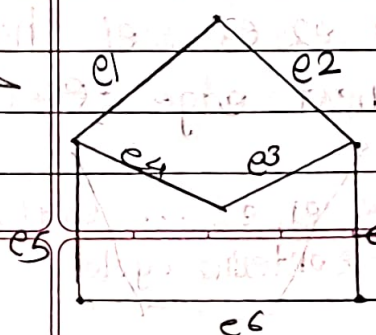
1. Tree is simple graph not contains loop as well as parallel edges.

2. Tree on n vertices has $(n-1)$ edges.

3. every tree contains atleast one pendant vertex

* Spanning Tree - let G be a connected graph. a subgraph (T) of G is said to be spanning tree if T is a tree and T passes through each vertex of G .

ex. find all spanning trees of the foll. graph.

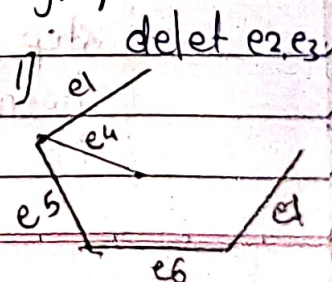


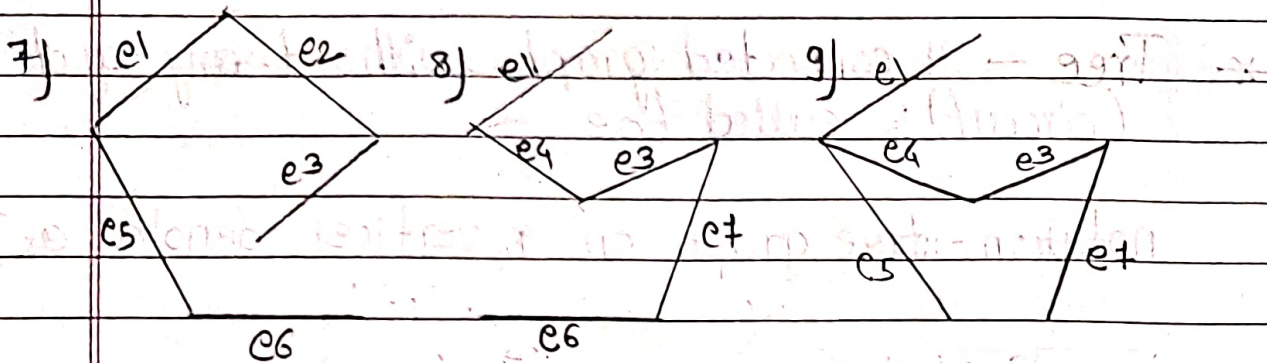
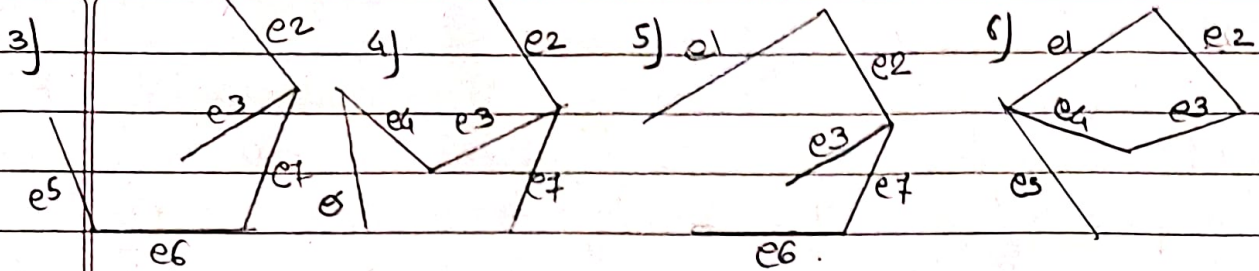
$\therefore$ no. of vertices $n = 6$

$\therefore$ no. of edges = 7

$\therefore$ edges of spanning tree

contains $(n-1) = 6-1 = 5$





* Shortest Spanning Tree

A spanning tree with minimum weight is called as shortest spanning tree.

* Kruskal's Algorithm -

Step 1:- Consider given connected ^{weighted} graph on n vertices.

Step 2:- find increasing sequence of weights of all edges of G . say $w_1, w_2, \dots, w_n$.

Step 3:- select edge e_i of minimum weight w_i which is not loop.

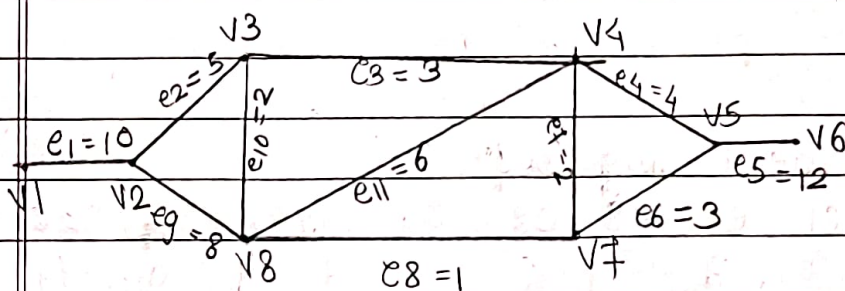
Step 4:- Suppose the edges $e_1, e_2, e_3, \dots, e_i$ have been chosen. then choose next edge e_{i+1} i.e. a) $e_{i+1} \neq e_k$
b) induced subgraph $G < e_1, e_2, \dots, e_{i+1} >$ not contains cycle.

c) $w(e_i+1)$ is of smallest weight satisfying a & b.

d) stop if $(n-1)$ edges have been chosen. else continue step four.

imp

1. use kruskal's algorithm find shortest spanning tree of the following graph. -6m



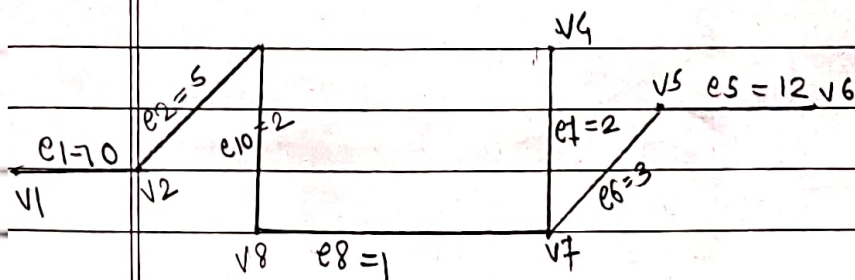
→ no. of vertices = 8

no. of edges = 11

edges	e1	e2	e3	e4	e5	e6	e7	e8	e9	e10	e11
weight	10	5	3	4	12	3	2	1	8	2	6

∴ by non-decreasing seq.

edges	e8	e7	e10	e6	e3	e4	e2	e11	e9	e1	e5
weight	1	2	2	3	3	4	5	6	8	10	12

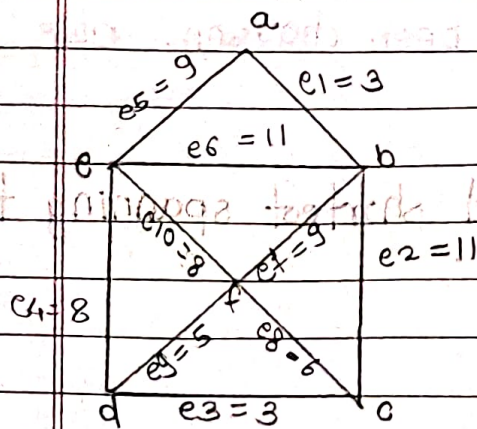


∴ weight of spanning tree is

$$= 10 + 5 + 2 + 8 + 1 + 3 + 2 + 12 + 10$$

$$= 35$$

1. using kruskal algorithm find shortest spanning tree in the following graph



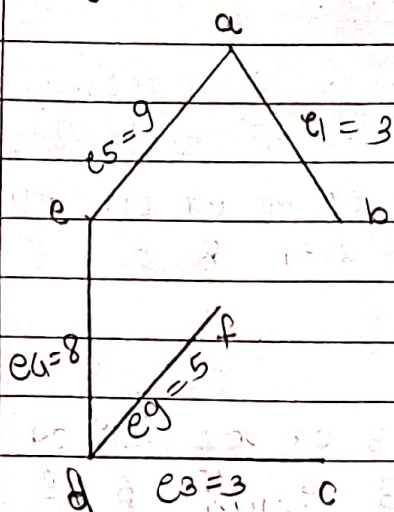
no. of vertices = 6

no. of edges = 10

edges	e1	e2	e3	e4	e5	e6	e7	e8	e9	e10
height	3	11	3	8	9	11	9	6	5	8

by non-decreasing seqⁿ.

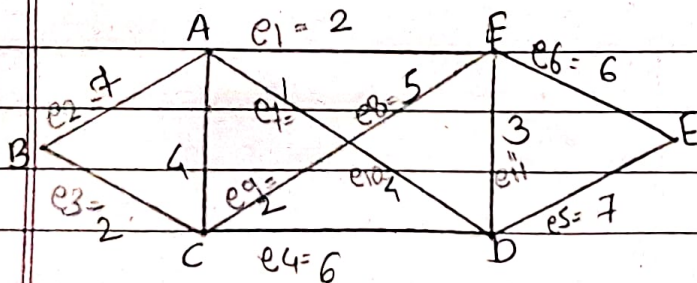
edges	e1	e3	e9	e8	e4	e10	e5	e7	e6	e2
height	3	3	5	6	8	8	9	9	11	11



∴ Total weight of spanning tree :-

$$9 + 3 + 8 + 5 + 3 = \underline{\underline{28}}$$

2.



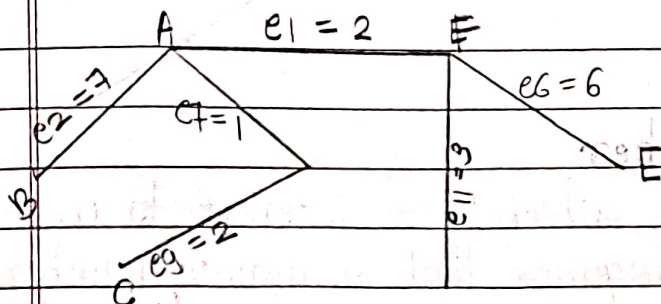
no. of vertices = 6

no. of edges = 11

edges	e1	e2	e3	e4	e5	e6	e7	e8	e9	e10	e11
height	2	7	2	6	7	6	1	5	2	4	3

by non-decreasing seqⁿ.

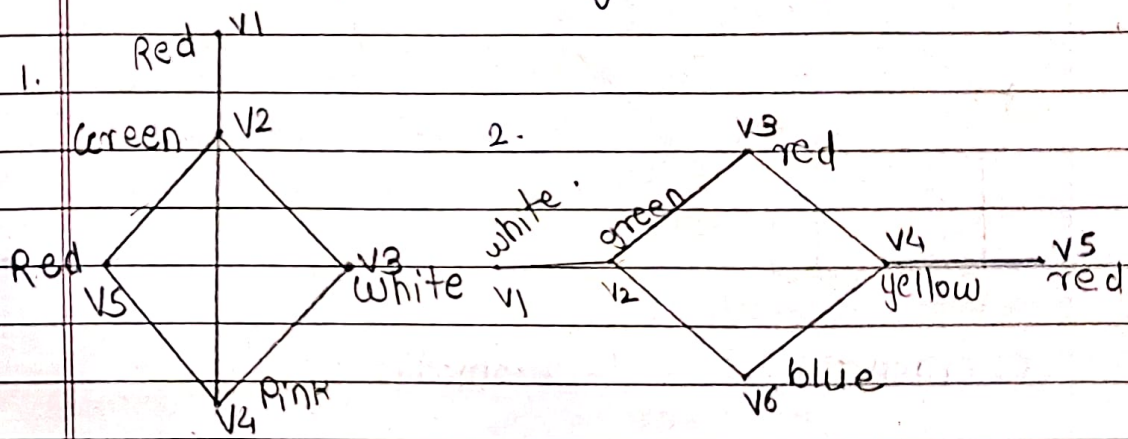
edges	e_7	e_1	e_3	e_9	e_{11}	e_{10}	e_8	e_4	e_6	e_5	e_2
height	1	2	2	2	3	4	5	6	6	7	7



total weight of tree :-
 $= 2 + 6 + 3 + 1 + 2 + 7$
 $= 21$

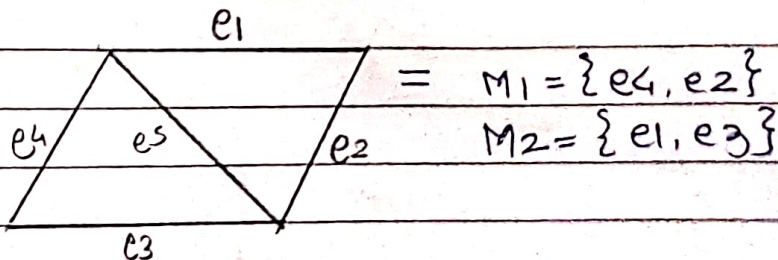
* Coloring of graph :-

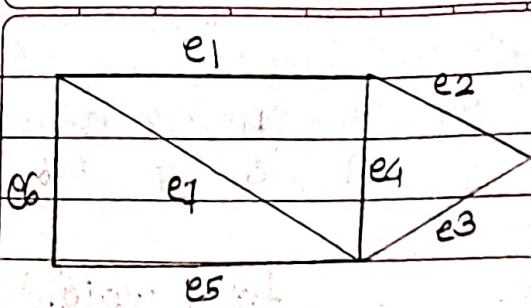
painting all vertices of a graph with colors such that no two adjacent vertices have the same color is called proper coloring to a graph.



* Matching :-

matching of a graph (G) is any subset subset of $(M \subseteq E)$ such that no two elements of M are adjacent
 for eg.





$$M_1 = \{e_1, e_5\}$$

$$M_2 = \{e_2, e_3\}$$

$$M_3 = \{e_6, e_4\}$$

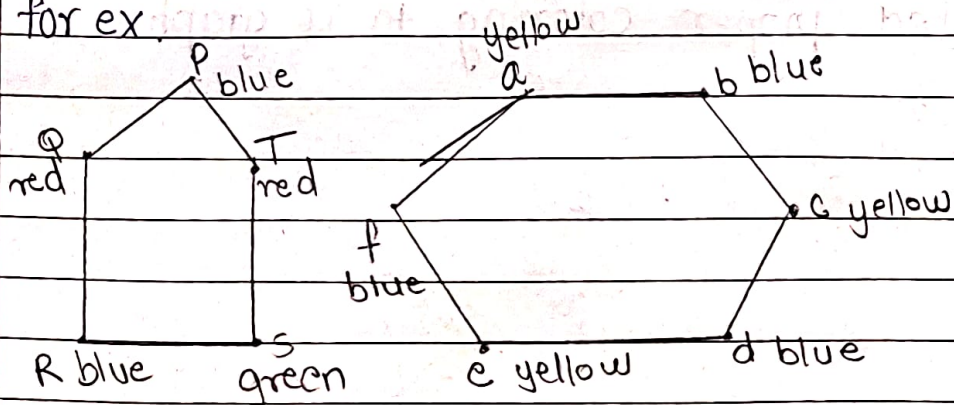
$$M_4 = \{e_2,$$

* Chromatic Number

proper coloring which is of interest to us is one of that requires that minimum number of color. the graph G that requires (k) different colors for its proper coloring is called k chromatic graph and the number k is called chromatic number of G .

for ex.

1.

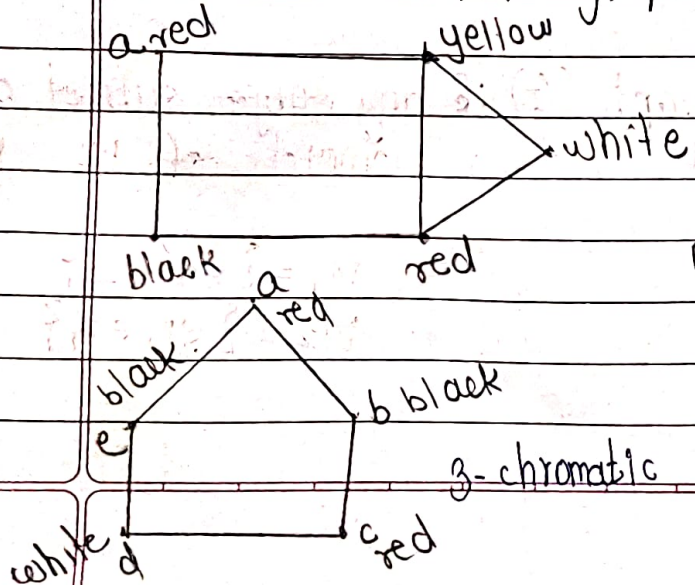


3 - chromatic

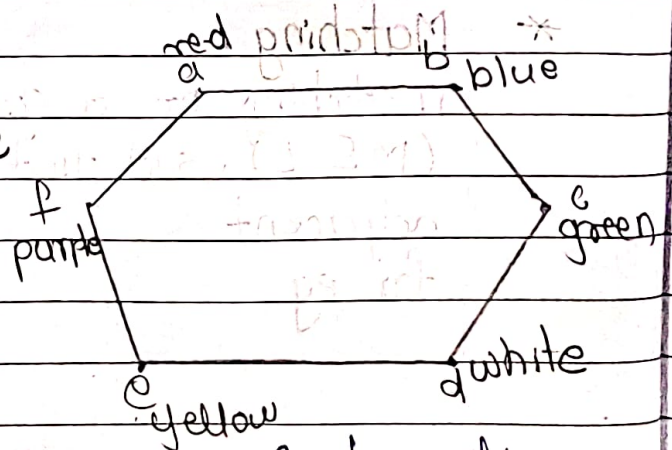
2 chromatic

2. draw 4 chromatic graph

3. draw 6 chromatic.

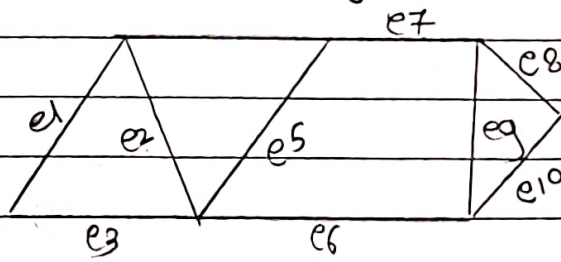


3 - chromatic



6 - chromatic

4. write down the matching of a following graph.



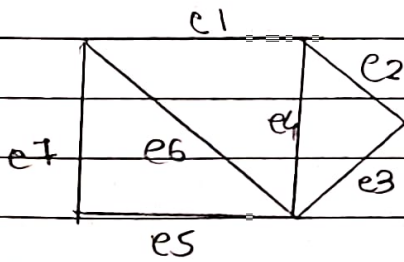
$$M_1 = \{e_7, e_6\}$$

$$M_2 = \{e_1, e_9\}$$

$$M_3 = \{e_2, e_5\}$$

$$M_4 = \{e_8, e_{10}\}$$

$$M_5 = \{e_7, e_3, e_6\}$$



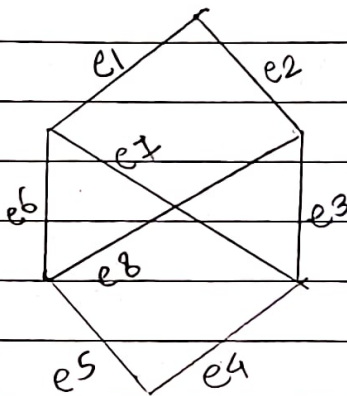
$$M_1 = \{e_1, e_5\}$$

$$M_2 = \{e_7, e_4\}$$

$$M_3 = \{e_2, e_3\}$$

$$M_4 = \{e_1, e_3\}$$

$$M_5 = \{e_2, e_5\}$$



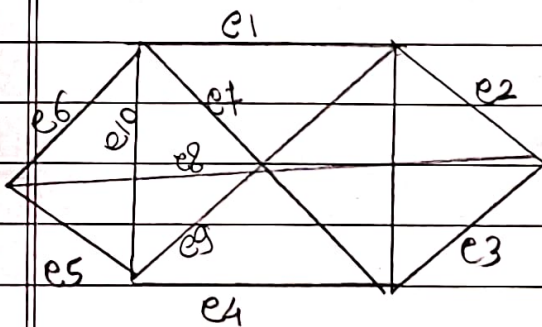
$$M_1 = \{e_1, e_3\}$$

$$M_2 = \{e_6, e_3\}$$

$$M_3 = \{e_3, e_5\}$$

$$M_4 = \{e_8, e_4\}$$

$$M_5 = \{e_4, e_6\}$$



$$M_1 = \{e_1, e_4\}$$

$$M_2 = \{e_8, e_4\}$$

$$M_3 = \{e_6, e_9\}$$

$$M_4 = \{e_5, e_1\}$$

$$M_5 = \{e_7, e_9\}$$

$$M_6 = \{e_2, e_3\}$$

$$M_7 = \{e_6, e_5\}$$